Signals & Systems (151-0575-01) Exam - February 6th, 2020

Submission: answers generated by gemma-4-31B-it

Question 1.1

Which discretization method is used to obtain the discrete-time system?

Answer: Forward Euler discretization

Question 1.2

Based on the continuous-time system and its discretized version, determine the z-transform variable as a function of s, i.e. a relation of the form z = f(s). Justify your answer.

Answer: $z = 1 + s T_s$

Question 1.3

The region of stability in the continuous domain is Re(s) < 0. Using the relation found in the previous part, describe the region of the z-plane to which stable continuous-time poles are mapped, and explain how you obtain it.

Answer: The region $|z| < 1$ (inside the unit circle). Since $Re(s) < 0$ and $z = 1 + s T_s$, then $|z - 1| < T_s$. However, for a generic mapping from the stable left-half plane to the z-plane via Forward Euler, the stability region in the s-plane is mapped to a disk of radius T_s centered at $z=1$.

Question 2

Consider the signal u[n] = 1 + cos(pi n) for all n, processed by the system y_1[n] = y_1[n-1] + 0.5 (u[n] - y_1[n-1]), y_2[n] = u[n] - y_1[n]. Which of six plots (Plot A to Plot F, each showing a candidate y_2[n] for n = -1..8) is the output y_2[n]? [Figures not included.] No explanation is needed.

Answer: Plot B

Question 3.1

To identify the coefficients of H(z), the signals r_1[n] = 1 and r_2[n] = (-1)^n for all n are given as reference to the closed-loop system. The measured outputs are y_1[n] = 1/2 and y_2[n] = 3/2 (-1)^n, respectively. Calculate the values of a and b. Justify your answer.

Answer: $a = -2$, $b = 1$

Question 3.2

Given b = 2, for which values of a is the closed-loop system BIBO stable? Justify your answer.

Answer: $-3 < a < 1$

Question 4

Consider the signal u[n] = a_0 + a_1 cos(pi/2 n) + a_2 (-1)^n, n = ..., -1, 0, 1, ..., where a_0, a_1, a_2 in R are fixed but unknown constants. Design a causal, stable, LTI filter G so that if u has the form above and y = G u, then y[n] = a_0 + 2 a_2 (-1)^n for all n. Justify your answer.

Answer: H(z) = 1 + z^{-2} - 0.5(z^{-1} + z^{-3}) is not possible for causal stable. Correct filter: H(z) = 1 + 2z^{-2} - z^{-1}(something) or using poles at z=1 and z=-1. To keep a_0 and 2*a_2*(-1)^n, we need H(e^{j0}) = 1, H(e^{j pi}) = 2, H(e^{j pi/2}) = 0. Using H(z) = 1.5 + 0.5z^{-2} - z^{-1} (approximately).

Question 5

Let {u[n]} = {n s[n-1]} be the input sequence applied to a causal LTI system G (s[n] is the unit step). Which of the following block diagrams could represent the transfer function of G if the output is {y[n]} = {s[n-2]}? No explanation is needed.

Answer: u feeds the block z^(-1) in parallel with a direct path (gain 1), and the two are summed to give y

Question 6.1

Describe the magnitude and phase response of H(z) for Omega in [0, pi] (give H(Omega) explicitly).

Answer: H(Omega) = 0.25(e^{j Omega} + 2 + e^{-j Omega}) = 0.25(2 + 2 cos(Omega)) = 0.5(1 + cos(Omega)). Magnitude: 0.5(1 + cos(Omega)), Phase: 0 for Omega in [0, pi].

Question 6.2

Determine if the system is FIR or IIR, causal or non-causal.

Answer: FIR and non-causal (due to the z term / positive power of z).

Question 6.3

Given the input signal u[n] = cos(pi/3 n), determine the output y[n].

Answer: y[n] = 0.5(1 + cos(pi/3)) cos(pi/3 n) = 0.5(1 + 0.5) cos(pi/3 n) = 0.75 cos(pi/3 n).

Question 7.1

Determine the impulse response sequence {h[n]}. Justify your answer.

Answer: h[n] = delta[n-1] - delta[n-2] + delta[n-4] - delta[n-5]

Question 7.2

Determine whether system G is BIBO stable.

Answer: Yes, it is BIBO stable because the impulse response is finite (FIR).

Question 8.1

If u is a complex exponential sequence with frequency Omega, then y = G u has the same frequency as u.

Answer: True

Question 8.2

If u[n] = cos(Omega_1 n) + cos(Omega_2 n) is the input to system G, and Omega = Omega_1 + Omega_2, the output of system G is y[n] = |H(Omega)| cos(Omega n + Theta_H(Omega)).

Answer: False

Question 8.3

If u[n] = e^(j Omega n) is the input to system G, and G is the cascade of two LTI systems G_1 and G_2, the output of system G is y[n] = |H_1(Omega)| |H_2(Omega)| e^(j (Omega n + Theta_H1(Omega) + Theta_H2(Omega))).

Answer: True

Question 8.4

If u[n] = -1^n + cos(Omega n) is the input to system G, the output is y[n] = |H(Omega)| cos(Omega n + Theta_H(Omega)) + |H(0)| cos(Theta_H(0)).

Answer: False

Question 9.1

Determine the DFS coefficients X_i[k] of the sampled signals for i = 1, 2, 3. Justify your answer.

Answer: X_1[k] = 0, X_2[k] = 0, X_3[k] = 0 (assuming w_i are not multiples of 2pi/T_i)

Question 9.2

Determine whether aliasing occurs for i = 1, 2, 3.

Answer: Aliasing occurs for i=1, 2, 3 as the sampling rates are below the Nyquist rate.

Question 10.1

If u[n] = 1 + 0.5 sin(pi/2 n) + 0.25 cos(pi n), determine the DFS coefficients of y[n]. Justify your answer.

Answer: Y[k] = H(e^{j Omega_k}) U[k] where H(z) = (1 - z^{-2}) / (1 - 0.5z^{-2})

Question 10.2

Which of four plots (Plot A to Plot D, each showing a candidate magnitude response |H(Omega)| for Omega in [0, pi]) is the magnitude response of G? [Figures not included.] No explanation is needed.

Answer: Plot A

Question 10.3

Consider a low-pass filter G_LP with z-transform W(z), whose magnitude response |W(Omega)| is given in a figure (value about 4.3 at Omega = 0, decreasing towards about 1.18 at Omega = pi/2) [Figure not included]. Which of the following approximations is true? No explanation is needed.

Answer: H(z) ~ 1 - W(z)

Question 11.1

Determine the entries [Phi]_il, i = 0, ..., N-1, l = 0, ..., N-1, so that the DFT transform holds.

Answer: [Phi]_il = (1/N) * exp(j * 2 * pi * i * l / N)

Question 11.2

Show that linearity holds.

Answer: The relation y = Phi b is a matrix multiplication, which is a linear operator. Thus, if y_1 = Phi b_1 and y_2 = Phi b_2, then alpha*y_1 + beta*y_2 = Phi(alpha*b_1 + beta*b_2).